

Algebra PhD Exam, May 1989

Do 7 out of the following questions.

1. This problem has four parts.
 - (a) Define: Solvable group.
 - (b) State the Burnside's theorem for solvability of a finite group.
 - (c) Let H be a normal subgroup of a group G . Prove that $C_G(H)$ is a normal subgroup of G .
 - (d) Let G be a finite group of order 12. Prove that G has a non-trivial normal 2-subgroup (i.e., there is a subgroup T of G such that $1 \neq T$, and T is a 2-subgroup, and T is normal in G).
2. Two permutation representations $\phi_i : G \rightarrow \Sigma(X_i)$, (the symmetric group on X_i), are equivalent if there exists $\alpha : X_1 \rightarrow X_2$ such that for all $g \in G$, $x \in X_1$, we have $x(g\phi_1)\alpha = x(\alpha)(g\phi_2)$. Prove or disprove that the following two permutation representations of S_4 of degree 6 are equivalent:
 - (a) on the right coset space of $\langle (12)(34), (13)(24) \rangle$.
 - (b) on the right coset space of $\langle (12), (34) \rangle$.
3. Let F_n be the free group on n generators $x_1, \dots, x_n$. Let F'_n denote the commutator subgroup. Prove that F_n/F'_n is the free abelian group on $\{x_1, \dots, x_n\}$.
4. In each of the following groups decide whether it is a semi-direct product of two of its proper subgroups: D_4 (the dihedral group of order 8); Q_8 (the quaternion group of order 8). Justify your answer.
5. Let A be a commutative ring with identity. Suppose that for each prime ideal P , the local ring A_P has no nilpotent element $\neq 0$. Show that A has no nilpotent element $\neq 0$. If each A_P is an integral domain, is A necessarily an integral domain? Justify your answer.
6. Let A be a commutative ring with identity, a an ideal, M an A -module.
 - (a) Prove that $A \otimes_A M \cong M$ as A -modules.
 - (b) Prove that $(A/a) \otimes_A M$ is isomorphic to M/aM .
7. Let $A \subseteq B$ be integral domains, B integral over A . Prove that B is a field if and only if A is a field.
8. Find a minimal primary decomposition for the following ideals. Also give the prime ideals to which the primary ideals belong.
 - (a) (36) as an ideal in $\mathbb{Z}$.
 - (b) (x^2, xy) as an ideal in $\mathbb{C}[x, y]$.
9. State the following theorems.
 - (a) Nakayama's lemma.
 - (b) Hilbert basis theorem.
 - (c) Jacobson's Density theorem.
 - (d) Maschke's theorem.
 - (e) Wedderburn's theorem for finite dimensional simple algebras (or Wedderburn–Artin's theorem for simple Artinian rings).